EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 07

Basic Maths and Calculus

Complete G.P. and Straight-Line Graphs

By - Excellence Programme

10 September 2026

Topics to be covered

01 Quick G.P. check - identify a sequence using its common ratio

02 Sum of the first n terms - formula and simple derivation

03 Infinite G.P. - meaning, condition $|r|<1$ and why the formula works

04 Finite-sum and infinite-sum questions with separate solutions

05 Physics applications of G.P.

06 Coordinates, distance, slope and straight-line graphs

Rough work / extra work

GP → Is a series eg no

$a_1 r^{n-1} = r^0 \quad a_2 = ar$

$A = a, ar, ar^2, ar^3 - - - ar^{n-1}$

first term

$r = \frac{a_2}{a_1} = \frac{a_4}{a_3} = \frac{a_n}{a_{n-1}}$

integers > fraction or Irrational

Rough work / extra work

$$a_n = a \times r^{(n-1)}$$

G.P

S_n = Sum of n terms of a G.P/AP

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

Rough work / extra work

$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \ldots\ldots \infty \downarrow 0$

$S_n = \frac{a(r^n - 1)}{r - 1}$

$r^n = r^\infty = 0$ put

$r < 1$

let assume $r = 0.5 \rightarrow$

$r^2 = 0.25 \longrightarrow 0$

$r^3 = 0.125$

$r^4 = 0.0625$

$r^5 = 0.03125 \longrightarrow r^\infty \longrightarrow 0$

$\frac{a(-1)}{r - 1} = S_n$

Rough work / extra work

$$a_n = a \times r^{(n-1)}$$

$a = 2$, $r = 4$

$a_3 = ? = 32$

$\Rightarrow 2 \times (4)^{3-1}$

$\Rightarrow 2 \times 4^2$

$\Rightarrow 2 \times 16$

$\Rightarrow$ **32**

Which sequences are G.P.?

Check the ratio of consecutive terms. Which sequences are G.P.?

A. 2, 4, 8, 12, 24, 48

B. 2, 4, 8, 16, 32, 64

C. 1, 1/2, 1/4, 1/8, ...

D. 1, -1/2, 1/4, -1/8, ...

Simple solution

sequence	ratio	answer
2,4,8,12,24,48	ratio changes	not G.P.
2,4,8,16,32,64	2	G.P.
1,1/2,1/4,1/8,...	1/2	G.P.
1,-1/2,1/4,-1/8,...	-1/2	G.P.

A sequence is G.P. only when every consecutive ratio is the same.

Sum of the first n terms of a G.P.

For a G.P. with first term a and common ratio r:

Sn = a + ar + ar^2 + ... + ar^(n-1)

ar^{n-1}

Sn = a(1-r^n)/(1-r), r not equal to 1

Sn = a(r^n - 1)/(r - 1), r not equal to 1

Both boxed forms are the same. Use the one that keeps the arithmetic simple.

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

where $r \neq 1$

Why the finite-sum formula works

$S_n = a + ar + ar^2 + \dots + ar^{n-1}$

$rS_n = ar + ar^2 + \dots + ar^n$

$S_n - rS_n = a - ar^n$

$S_n(1-r) = a(1-r^n)$

Therefore $S_n = a(1-r^n)/(1-r)$

Special case: r = 1

If r=1, every term is equal to a.

a + a + a + ... (n terms)

When r=1: Sn = na

Do not put r=1 into a(1-r^n)/(1-r), because it gives 0/0.

$r = 1$

$a, ar, ar^2, ar^3 \ldots ar^{n-1}$

If $r = 1$

$a, a(1)^1, a(1)^2, a(1)^3 \ldots a(1)^{n-1} \to 1$

$a, a, a, a, a \ldots a = n$

n times

Finite G.P.: first five terms

Find the sum of the first 5 terms: 2 + 6 + 18 + 54 + ...

$S_5 = 242$

$a = 2$
$r = 3$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{2 \times (3^5 - 1)}{(3 - 1)}$$

$$= 3^5 - 1 = 243 - 1$$

$$= 242$$

Simple solution

Step 1: a=2, r=6/2=3, n=5

Step 2: S5 = 2(3^5-1)/(3-1)

Step 3: S5 = 2(243-1)/2 = 242

Sum of the first 5 terms = 242

Finite G.P.: ratio smaller than one

Find the sum of the first 6 terms: 3 + 3/2 + 3/4 + 3/8 + ...

$a_6 = ?$ $\quad 3/32$

$S_6 = ?$ $\quad \frac{189}{32}$ Good

$r = \frac{1}{2}, \; a = 3$

$a_n = a r^{n-1}$

$= 3 \times \left(\frac{1}{2}\right)^5 \Rightarrow \frac{3}{32}$

$S_6 = \frac{a(r^n - 1)}{r - 1}$

$= \frac{3\left(\left(\frac{1}{2}\right)^6 - 1\right)}{\frac{1}{2} - 1}$

$= \frac{3\left(\frac{1}{64} - 1\right)}{-\frac{1}{2}}$

Rough work / extra work

$$\frac{3\left(\frac{1}{64} - 1\right)}{-\frac{1}{2}}$$

$$\frac{3\left(\frac{1-64}{64}\right) \times (-2)}{\frac{3\times -63\times -2}{64\ 32}}$$

$$\Rightarrow \frac{189}{32}$$

Simple solution

Step 1: a=3, r=1/2, n=6

Step 2: S6 = 3[1-(1/2)^6]/[1-1/2]

Step 3: S6 = 3(63/64)/(1/2) = 189/32

S6 = 189/32 = 5.90625

What does an infinite G.P. mean?

An infinite G.P. keeps adding terms forever.

a + ar + ar^2 + ar^3 + ...

It can have a finite sum only when the terms become smaller and move towards zero.

A finite infinite-sum needs |r|<1

$n = \infty$

$r < 1$

$r^n = r^\infty = 0$

$S_n = a\left(\frac{r^n - 1}{r - 1}\right)$ → 0

$= a\left(\frac{-1}{r - 1}\right)$

Why |r| must be less than 1

value of r	what happens to r^n	infinite sum
|r|<1	r^n approaches 0	finite / converges
r=1	stays 1	keeps growing
r=-1	changes +1, -1	oscillates
|r|>1	magnitude grows	diverges

$S_n = na$

Terms going on forever does not always mean that a finite sum exists.

$r = -1$

$r^2 = 1$

$r^3 = -1$

$\rightarrow -a, \; a, \; -a, \; -a \; \ldots \; a(-1)^n$

From finite sum to infinite sum

Sn = a(1-r^n)/(1-r)

When |r|<1 and n becomes very large, r^n becomes nearly zero.

S(infinity) = a(1-0)/(1-r)

S(infinity) = a/(1-r), only when |r|<1

$$S_\infty = \frac{a}{1-r}$$

Half + quarter + eighth fills one whole

Standard Value

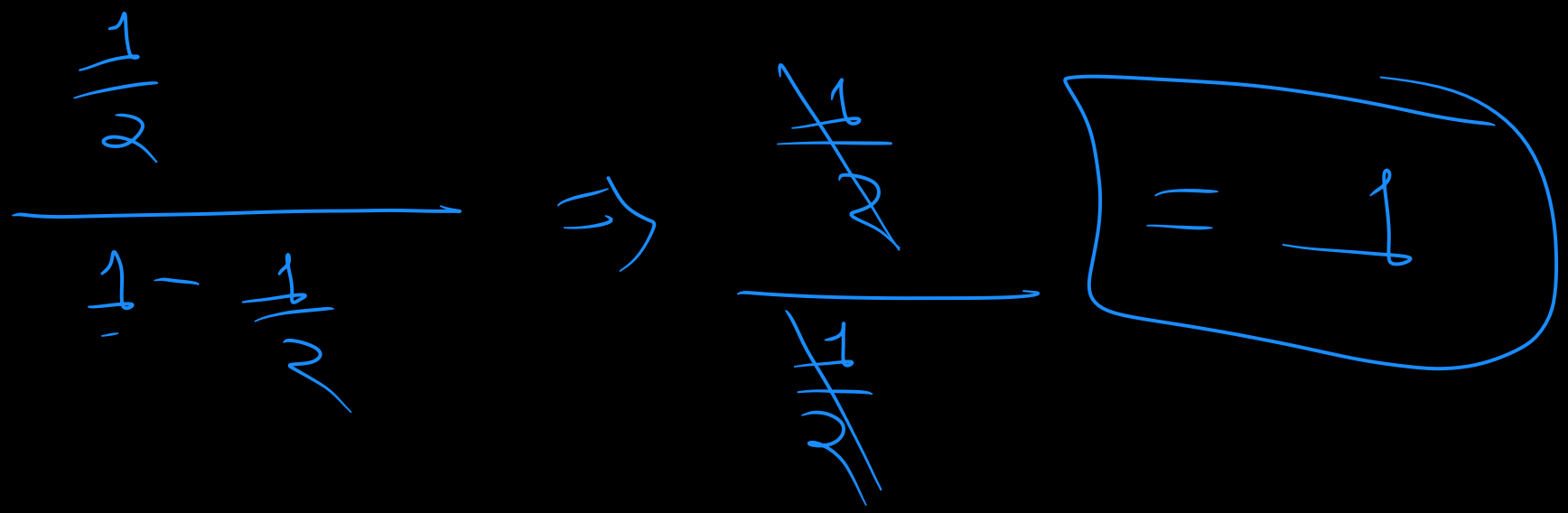
Imagine filling one rectangle in smaller parts:

1/2 + 1/4 + 1/8 + 1/16 + ...

a=1/2, r=1/2

S(infinity) = (1/2)/(1-1/2) = 1

The pieces become tiny, but together they fill exactly 1 whole.



Rough work / extra work

Finite or infinite? Use this checklist

1. Is the number of terms given? Use Sn.
2. Do the dots continue forever? It may be an infinite series.
3. For an infinite series, calculate r.
4. Use S(infinity) only if |r|<1.

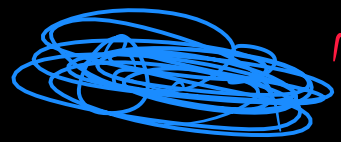
$$S_\infty = \frac{a}{1-r}$$ only if $r < 1$

Never use the infinite-sum formula when |r| is 1 or more.

$$r^\infty = 0$$

Infinite G.P.: 1 + 1/3 + 1/9 + ...

Find 1 + 1/3 + 1/9 + 1/27 + ...

$S_\infty =$ 

$r = \frac{\frac{1}{3}}{1} = \frac{1}{3}$

$a = 1$

$S_\infty = \frac{a}{1-r} = \frac{1}{1-\frac{1}{3}} = \frac{1}{\frac{2}{3}}$

$\Rightarrow \frac{3}{2}$

Simple solution

Step 1: $a=1 \qquad r=1/3 \qquad |r|<1$

Step 2: $S_\infty = a/(1-r) = 1/(1-1/3) = 1/(2/3)$

$S_\infty = 3/2$

Potential at origin — alternating charges

+q, −q, +q, … are at x=1,2,4,8,… Find potential at origin.

Ion → Charge (+, −) →

q_1 q_2 q_3 − − − q_∞

$+q, -q, +q \ldots$

x_1 x_2 x_3 − − − x_∞

$x = 1, 2, 4, 8 \ldots$

$$\text{Total charge} = \sum_{i=1}^{\infty} (x_i \cdot q_i)$$

Infinite resistance network

Equivalent series blocks are R, R/2, R/4, R/8, ... Find R_AB.

$$T = 1 \times q + 2 \times (-q) + 4 \times q + 8(-q) - \cdots \text{(1)}$$

It is a G.P

$a = ? = q$

$$r = \frac{2 \times (-q)}{q} = -2$$

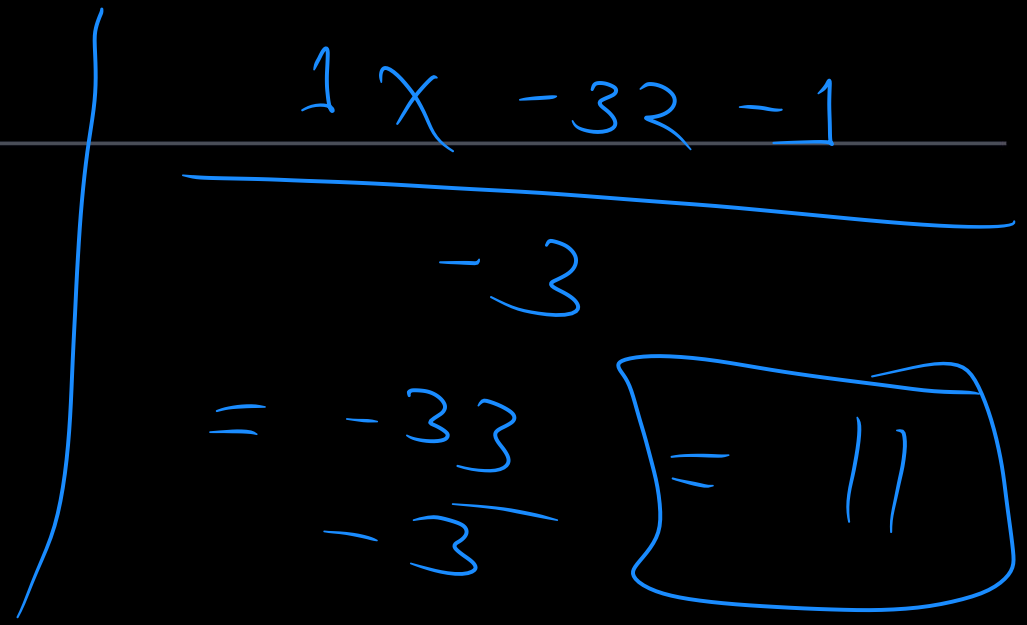
$$T = q\left(1 + -2 + 4 + -8 + 16\right) = \frac{q \times \left((-2)^5 - 1\right)}{-2 - 1}$$

Simple solution

Step 1: $V = kq\,[1 - 1/2 + 1/4 - 1/8 + \ldots]$

Step 2: $a=1 \qquad r=-1/2 \qquad S_\infty = 1/[1-(-1/2)] = 2/3$

$V = 2kq/3$



Rough work / extra work

Simple solution

Step 1: $R_{AB} = R\,[1+1/2+1/4+1/8+\ldots]$

Step 2: $R_{AB} = R \times 1/(1-1/2)$

$$R_{AB} = 2R$$

Force on q_0 at the origin

Charges q lie at x=1,2,4,8,... . Find force on q_0 at origin.

Simple solution

Step 1: $F = kqq_0 [1/1^2 + 1/2^2 + 1/4^2 + \ldots]$

Step 2: $F = kqq_0 [1 + 1/4 + 1/16 + \ldots] = kqq_0/(1-1/4)$

$|F| = 4kqq_0/3$ (toward $-x$ for like charges)

Fifth resistance when each is half

Five resistors are in series. Each resistance is half the previous. First is 10 Ω. Find the fifth.

Simple solution

Step 1: $a = 10\ \Omega$ $r = 1/2$ $T_5 = ar^4$

Step 2: $R_5 = 10(1/2)^4 = 10/16$

$R_5 = 5/8\ \Omega = 0.625\ \Omega$

Rough work / extra work

Choose the correct G.P. formula

G.P

one term: Tn = ar^(n-1) → $T_n = ar^{n-1}$

first n terms: Sn = a(1-r^n)/(1-r), r not equal to 1 → $S_n = \frac{a(r^n - 1)}{r - 1}$

infinitely many terms: S(infinity) = a/(1-r), only when |r|<1 → $S_\infty \text{ & } |r|<1 = \frac{a}{1-r}$

First count the terms. Do not use the infinite-sum formula for a finite number of terms.

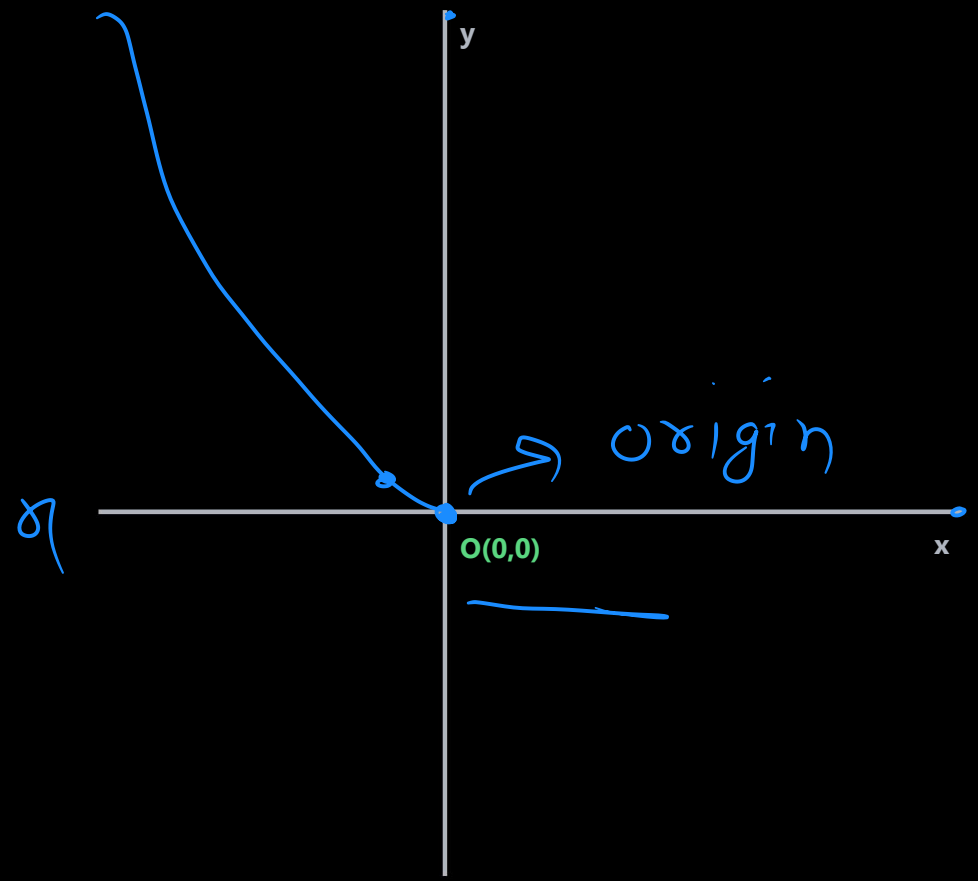
Graph — a picture of relation

A graph is a diagram in which a line or curve shows the relation between two physical quantities.

x-axis: horizontal **y-axis: vertical**

origin: O(0,0)

Always write the physical quantity and its unit on each axis.

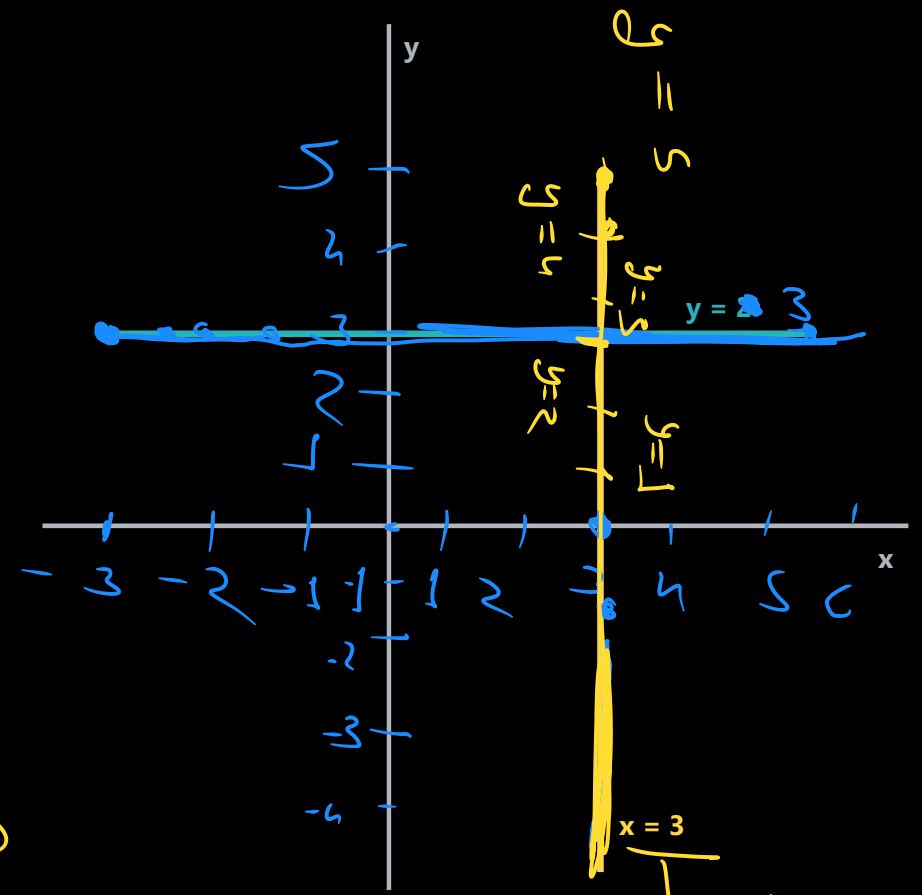


Constant lines — do not swap them

y = 2 → **horizontal line**
x changes but y stays 2.

x = 3 → **vertical line**
y changes but x stays 3.

$y = 3$

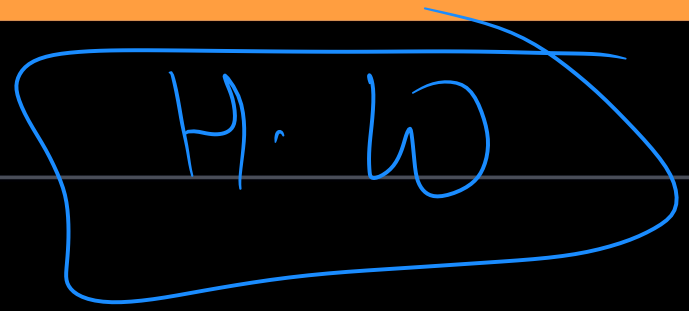


x is fixed & y is variable
line parallel to y-axis

y → fixed line ||to x-axis

↳ x = 3 line

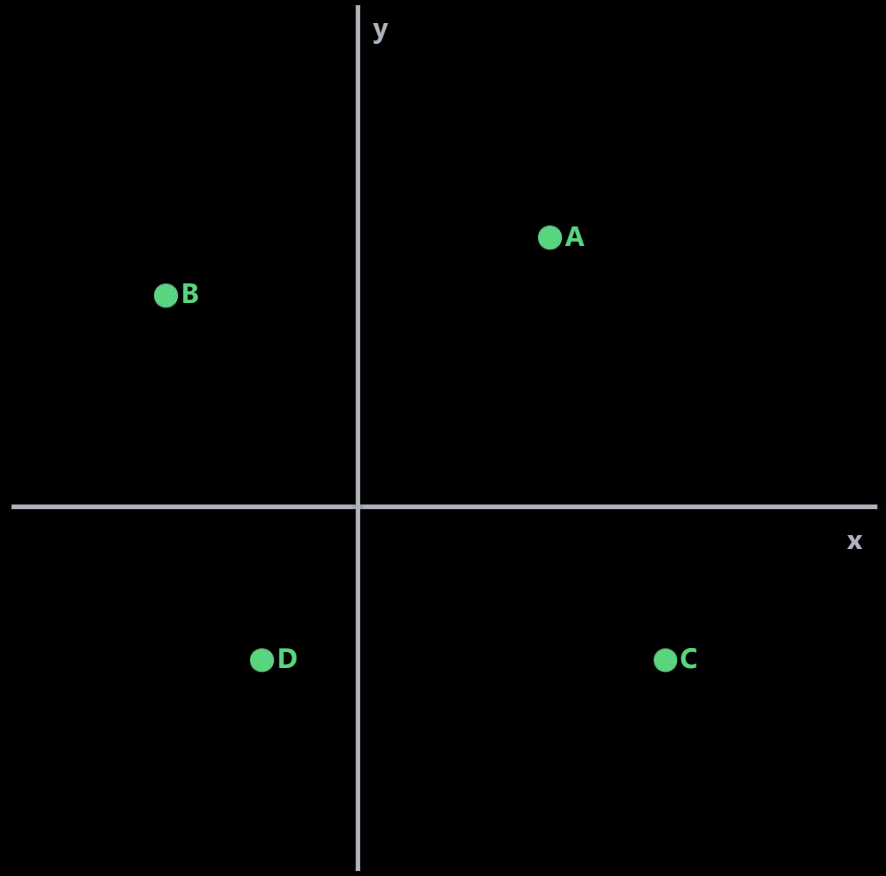
Locate four points



Locate A(2,4), B(−2,3), C(4,−2), D(−1,−2).

Simple solution

point	x sign	y sign	quadrant
A(2,4)	+	+	I
B(−2,3)	−	+	II
C(4,−2)	+	−	IV
D(−1,−2)	−	−	III



Rough work / extra work

Distance formula — two dimensions

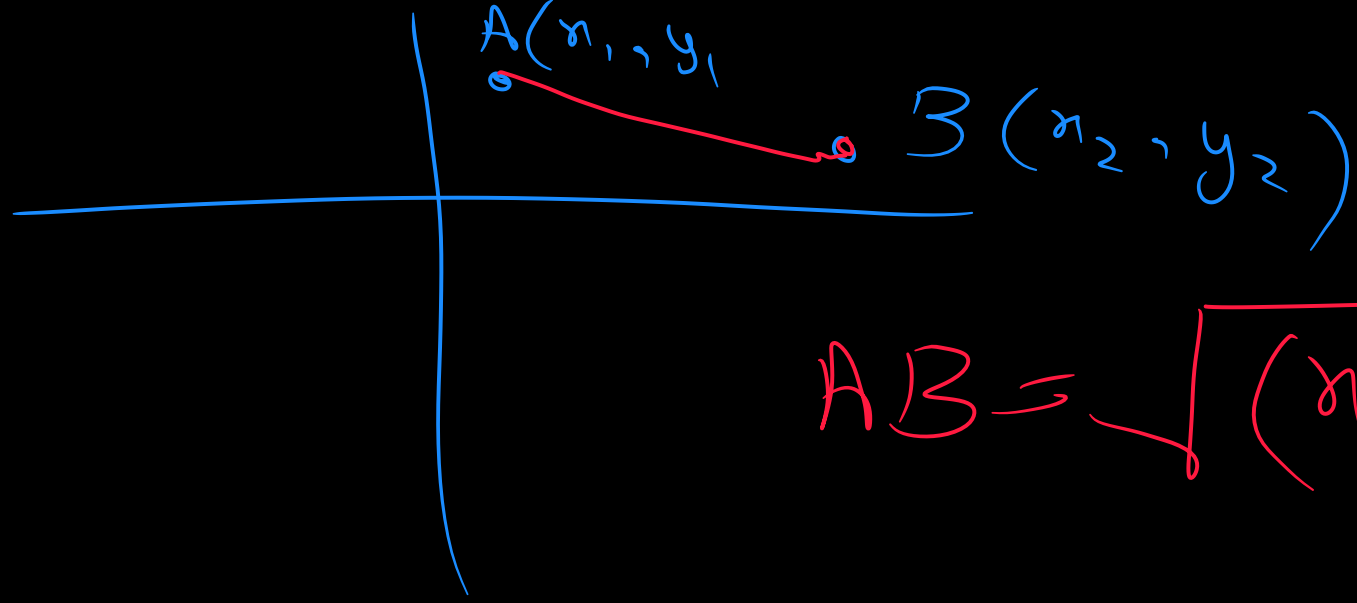
VIP

For $P_1(x_1,y_1)$ and $P_2(x_2,y_2)$, make a right triangle using horizontal and vertical changes.

$\Delta x = x_2 - x_1$ $\Delta y = y_2 - y_1$

$$P_1P_2 = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$$

Keep each difference inside a bracket before squaring.



$$AB = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

Distance in three dimensions

z_1 z_2

Find distance between P(2,−1,0) and Q(3,1,2).

$$= \sqrt{(3-2)^2 + (1-(-1))^2 + (2-0)^2}$$

$$= \sqrt{1^2 + 2^2 + 2^2}$$

$$= \sqrt{5+4}$$

$$= \sqrt{9} = 3$$

Simple solution

$$PQ = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]}$$

Step 1: $PQ = \sqrt{[(3-2)^2 + (1+1)^2 + (2-0)^2]}$

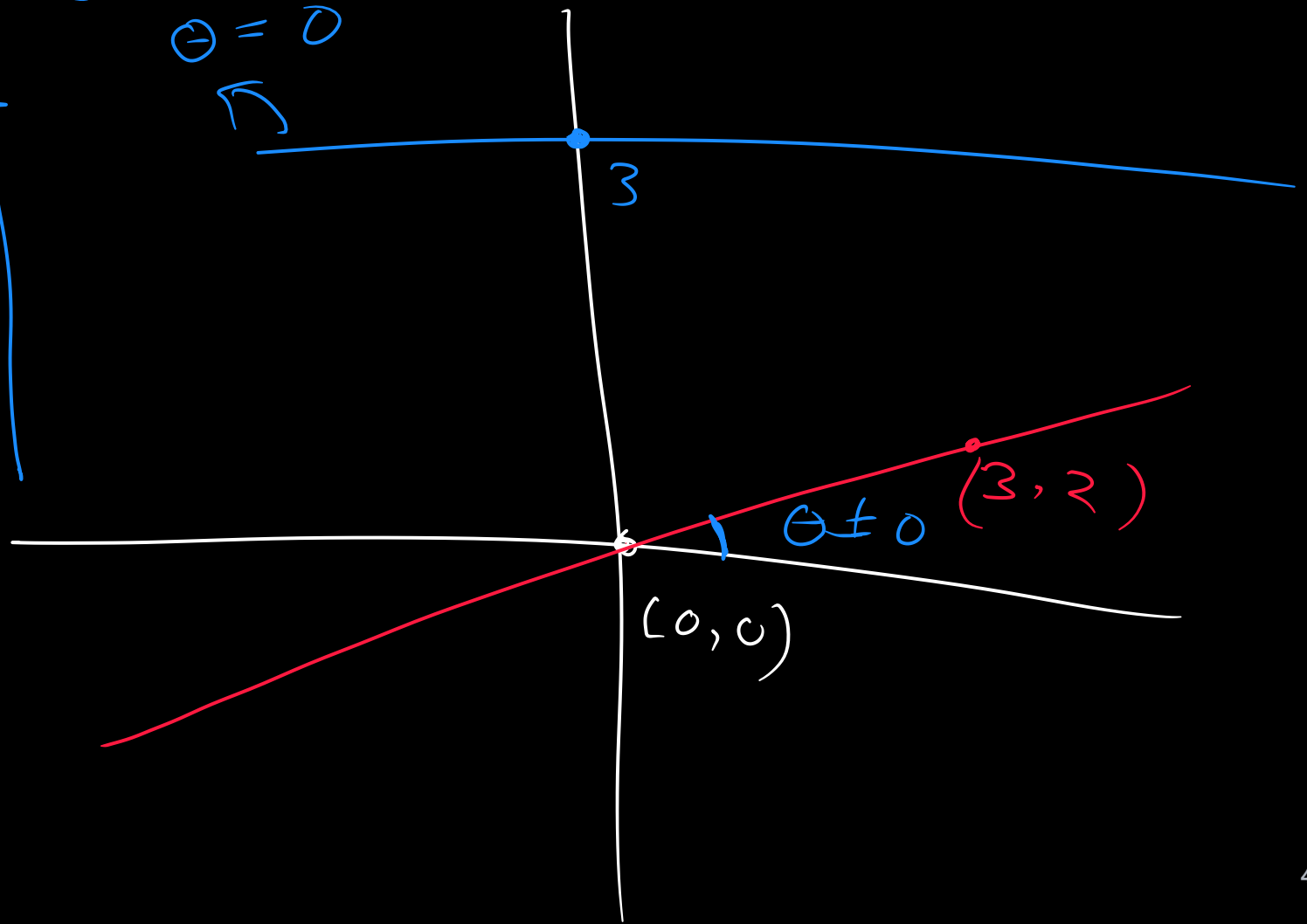
Step 2: $PQ = \sqrt{(1 + 4 + 4)} = \sqrt{9} = 3$

Distance = 3 units

Rough work / extra work

Equation $\Rightarrow$ $y = mx + c$

This tilt from x-axis is known as Slope of a line

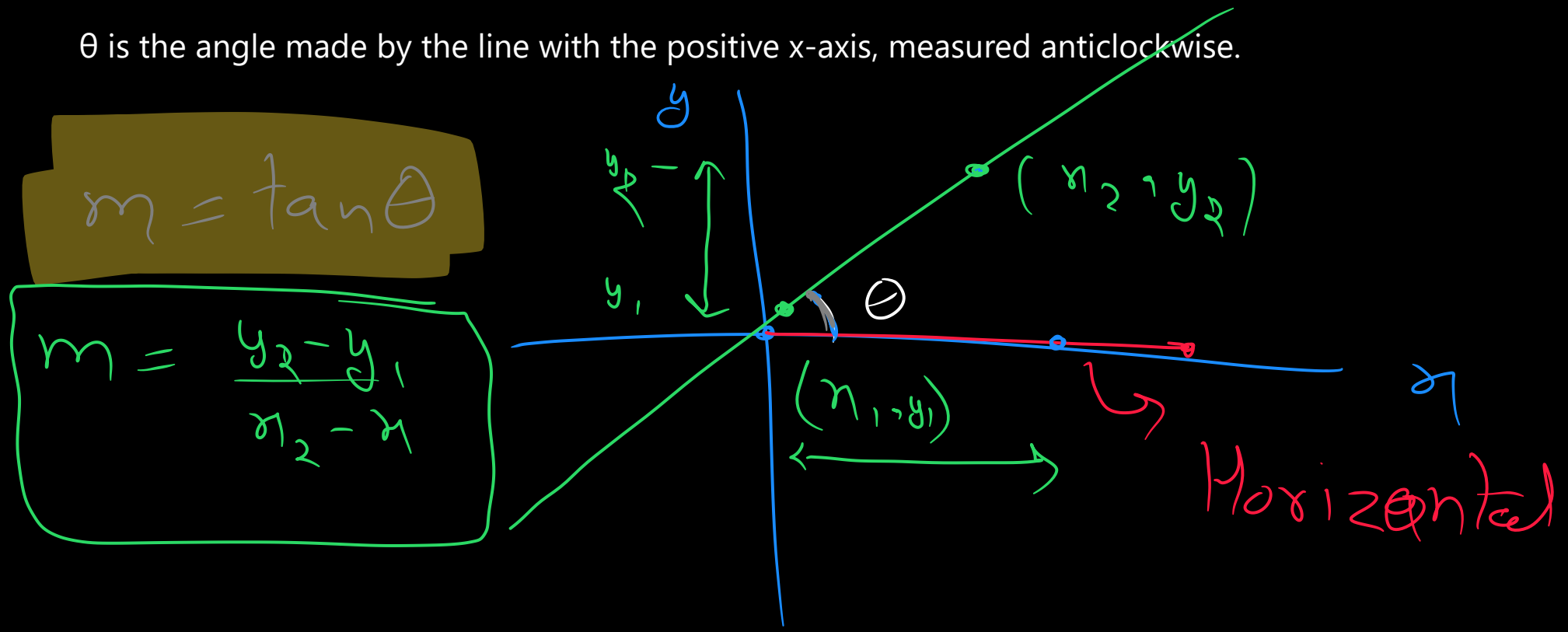


Slope — rise divided by run

Slope tells how quickly y changes when x changes.

$$m = \text{slope} = \tan\theta = \Delta y/\Delta x = (y_2 - y_1)/(x_2 - x_1)$$

θ is the angle made by the line with the positive x-axis, measured anticlockwise.



A straight line has the same slope everywhere

Choose any two points A and B on one straight line.

m_AB = (y_B−y_A)/(x_B−x_A)

Choose another pair C and D. The answer is unchanged.

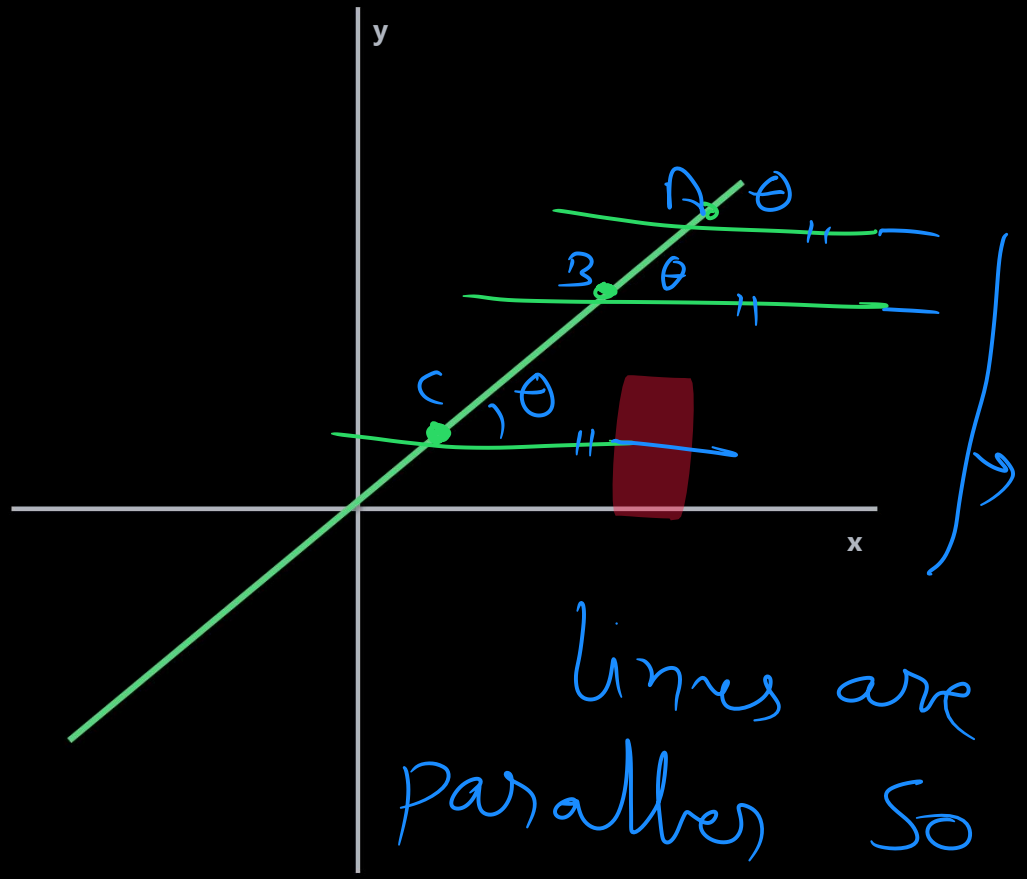
m_AB = m_CD

This is special to a straight line. A curve generally has changing slope.

$m_A = \tan\theta$

$m_B = \tan\theta$

$m_C = \tan\theta$



lines are parallel so angle will always same

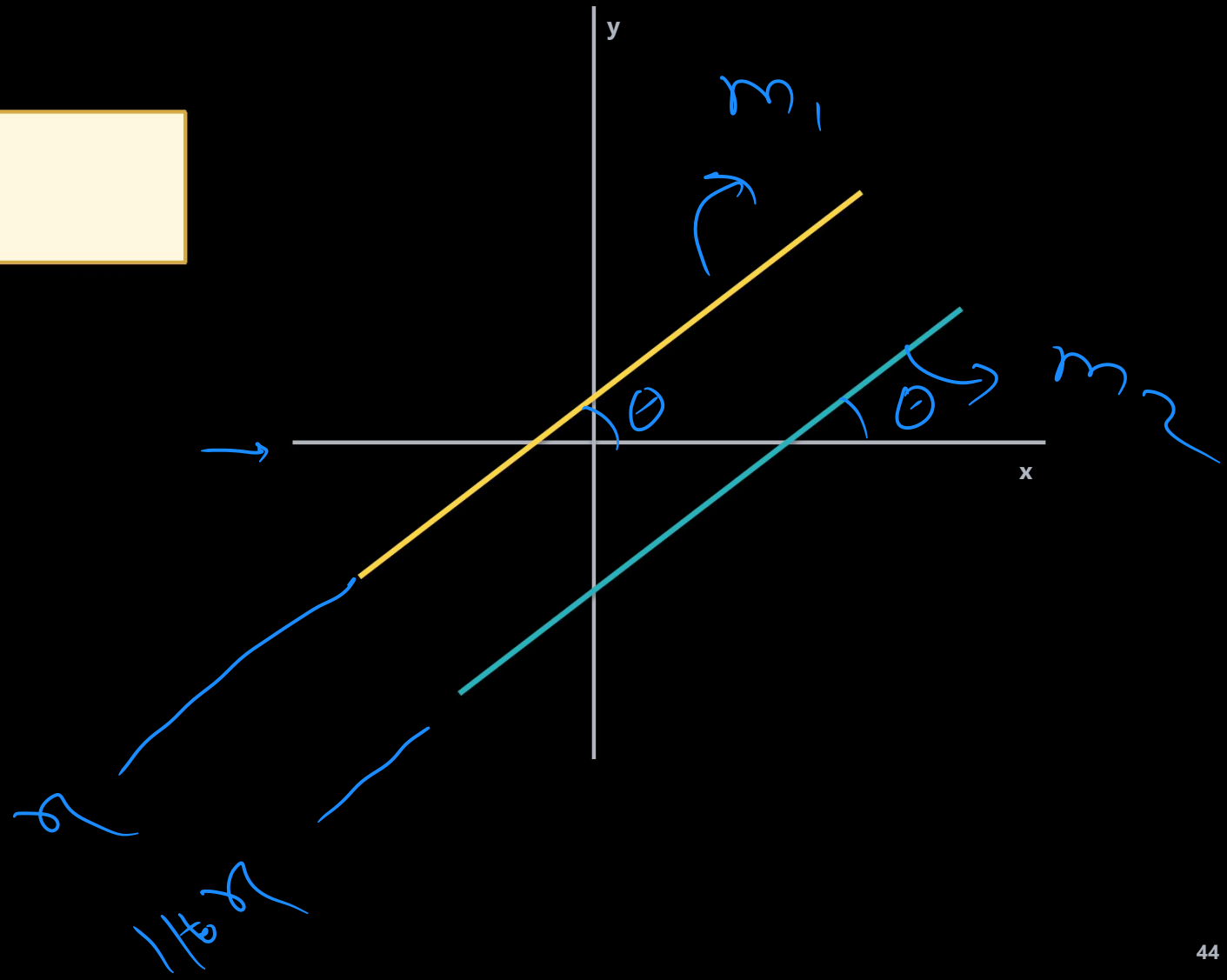
Parallel straight lines have equal slope

Parallel lines make the same angle θ with the positive x-axis.

$m_1 = \tan\theta$ $m_2 = \tan\theta$

parallel lines → $m_1 = m_2$

If 2 lines are parrel then this slope is Equal



Slope values you must know

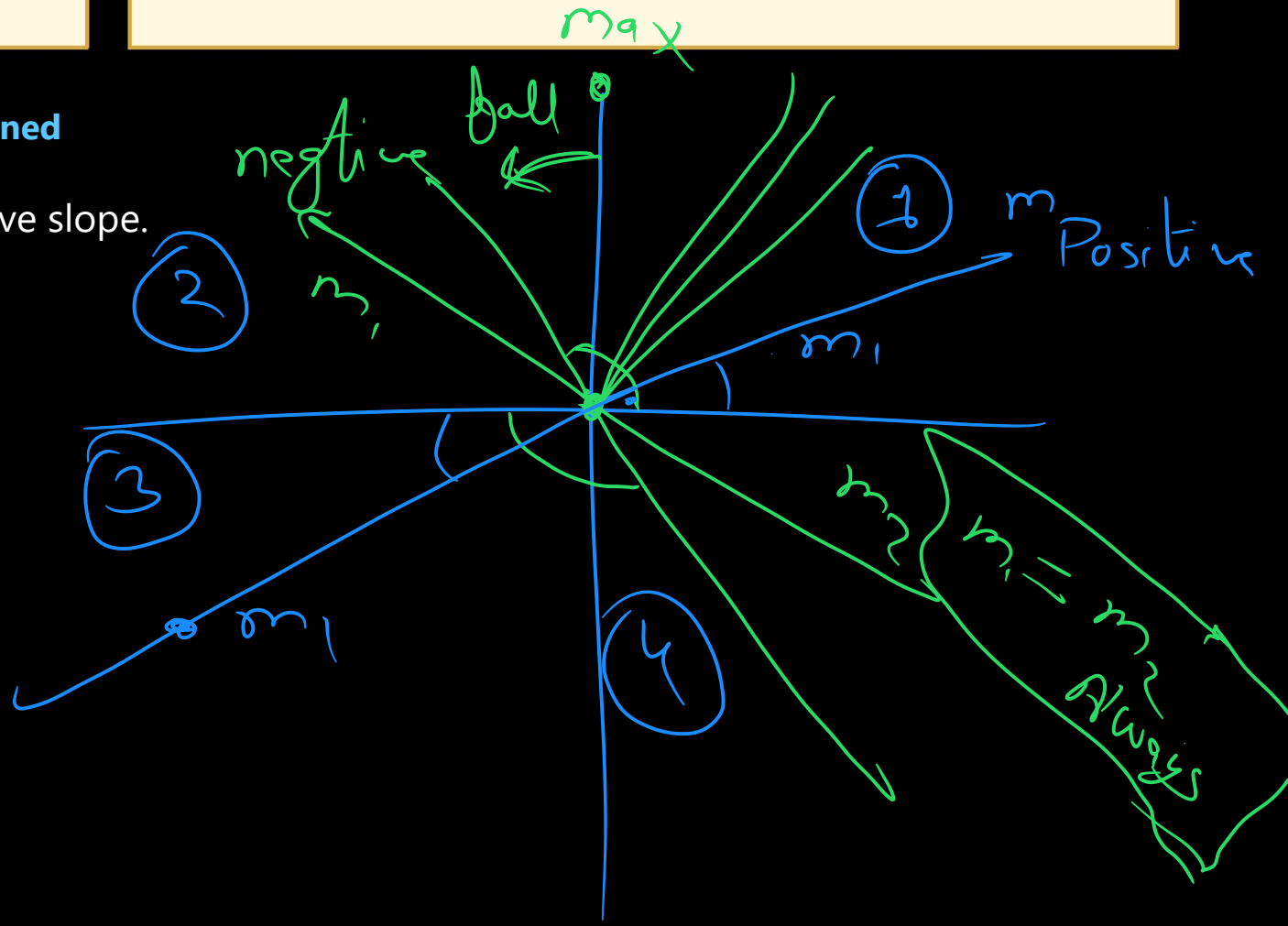
angle θ	slope tan θ	line feel
0°	0	horizontal
30°	1/√3	gentle rise
37°	3/4	rise
53°	4/3	steeper rise
90°	not defined / infinite	vertical
120°	−√3	falls
150°	−1/√3	gentle fall

Sign of slope from the angle → Sign of tan θ (A S T C)

$0° < \theta < 90°$ → $m > 0$	$90° < \theta < 180°$ → $m < 0$

$\theta = 0°$ → $m = 0$ **$\theta = 90°$ → vertical slope is not defined**

A rising line has positive slope. A falling line has negative slope.



trap — slope is not the angle

"The angle made by a straight line with +x-axis is called slope." True or false?

Angle = θ

but slope = $\tan\theta$ ⇒ It is Ratio not angle

Simple solution

Step 1: angle = θ
Step 2: slope = tan θ

False — slope is tan θ, not θ.

$$m = \text{Ratio} = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x}$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Rough work / extra work

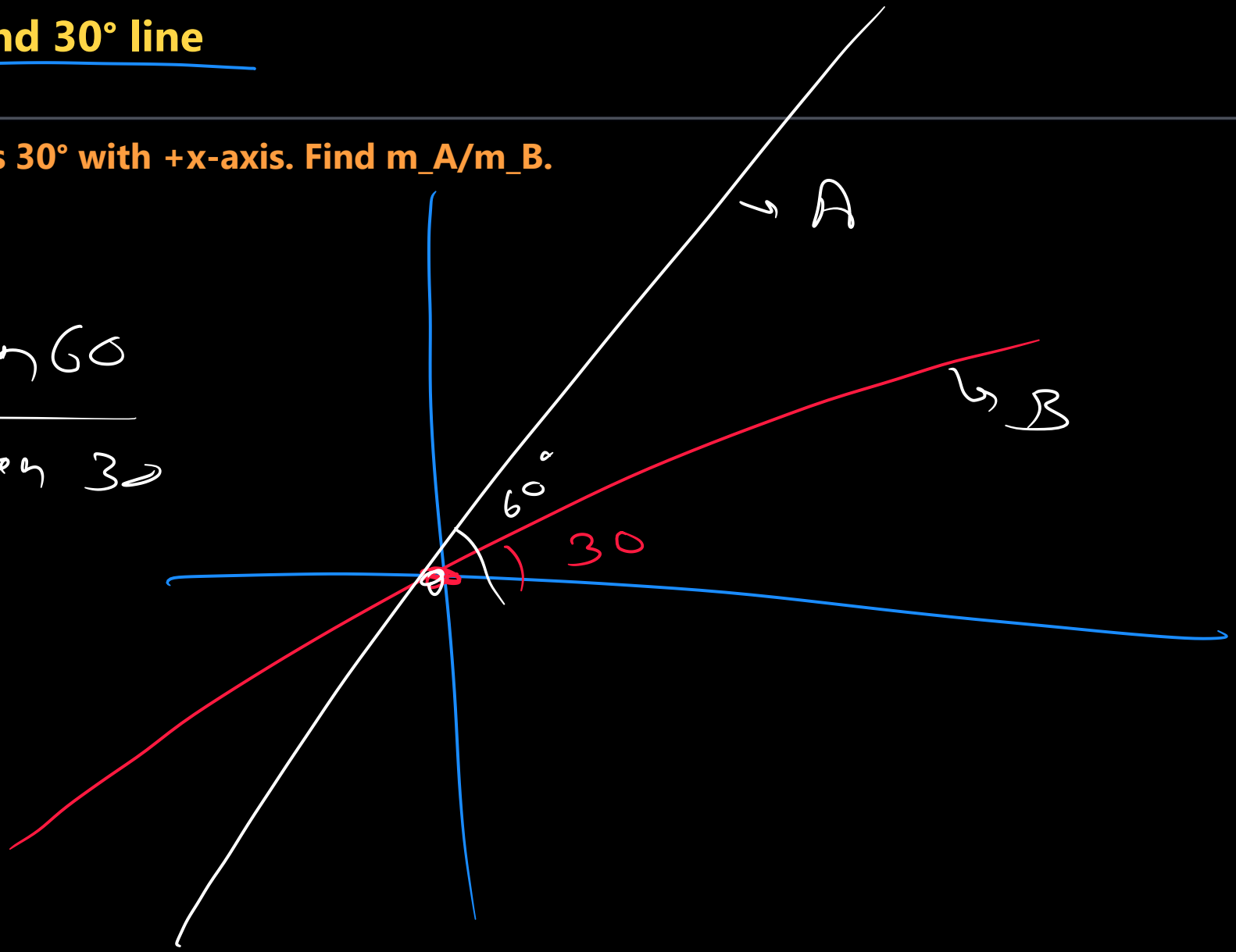
Ratio of slopes — 60° line and 30° line

Line A makes 60° and line B makes 30° with +x-axis. Find m_A/m_B.

$$\frac{m_A}{m_B} = \frac{?}{0} = \frac{\tan 60}{\tan 30}$$

$$= \frac{\sqrt{3}}{1/\sqrt{3}}$$

$$= 3/1$$



Simple solution

Step 1: $m_A/m_B = \tan60°/\tan30°$
Step 2: $m_A/m_B = \sqrt{3} \div (1/\sqrt{3}) = 3$

$m_A : m_B = 3 : 1$

Ratio of slopes — 30° line and 60° line

Now line A makes 30° and line B makes 60°. Find m_A/m_B.

Simple solution

Step 1: $m_A/m_B = \tan30°/\tan60°$
Step 2: $m_A/m_B = (1/\sqrt{3}) \div \sqrt{3} = 1/3$

$m_A : m_B = 1 : 3$

Rough work / extra work

Equation of a straight line

Every non-vertical straight line can be written in slope-intercept form.

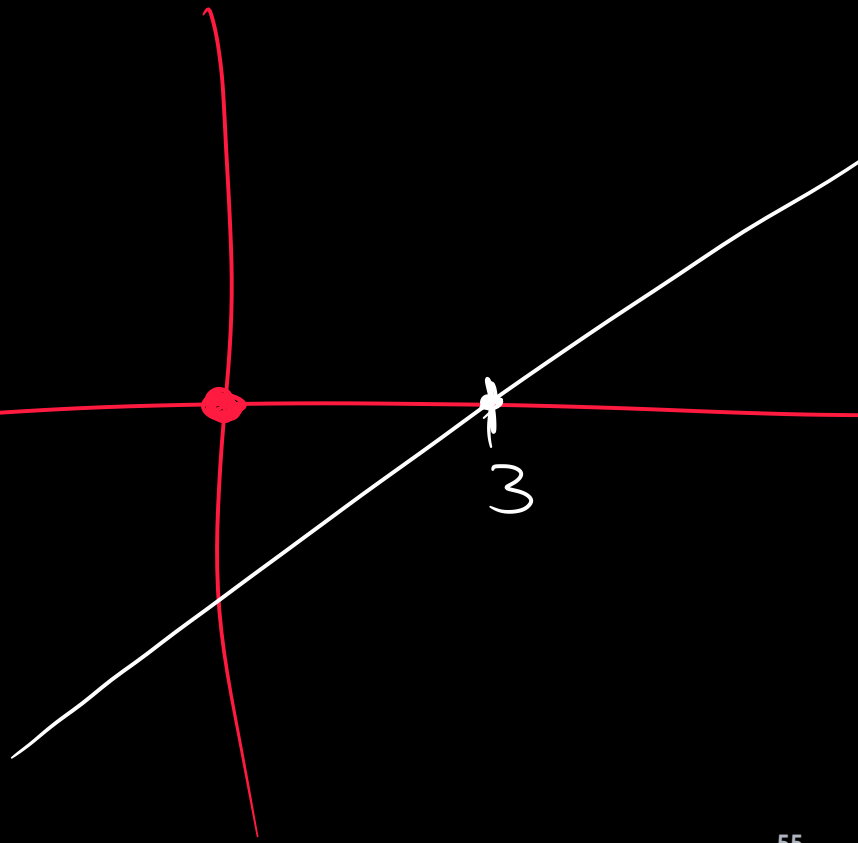
$$y = mx + c$$

m is slope of line

m = slope **c = y-intercept = value of y when x=0**

m and c are fixed for one straight line. x and y change from point to point.

If line don't pass from origin then distance point Intersection of line with x-axis & origin is called X intercept



Equation of a straight line

Every non-vertical straight line can be written in slope-intercept form.

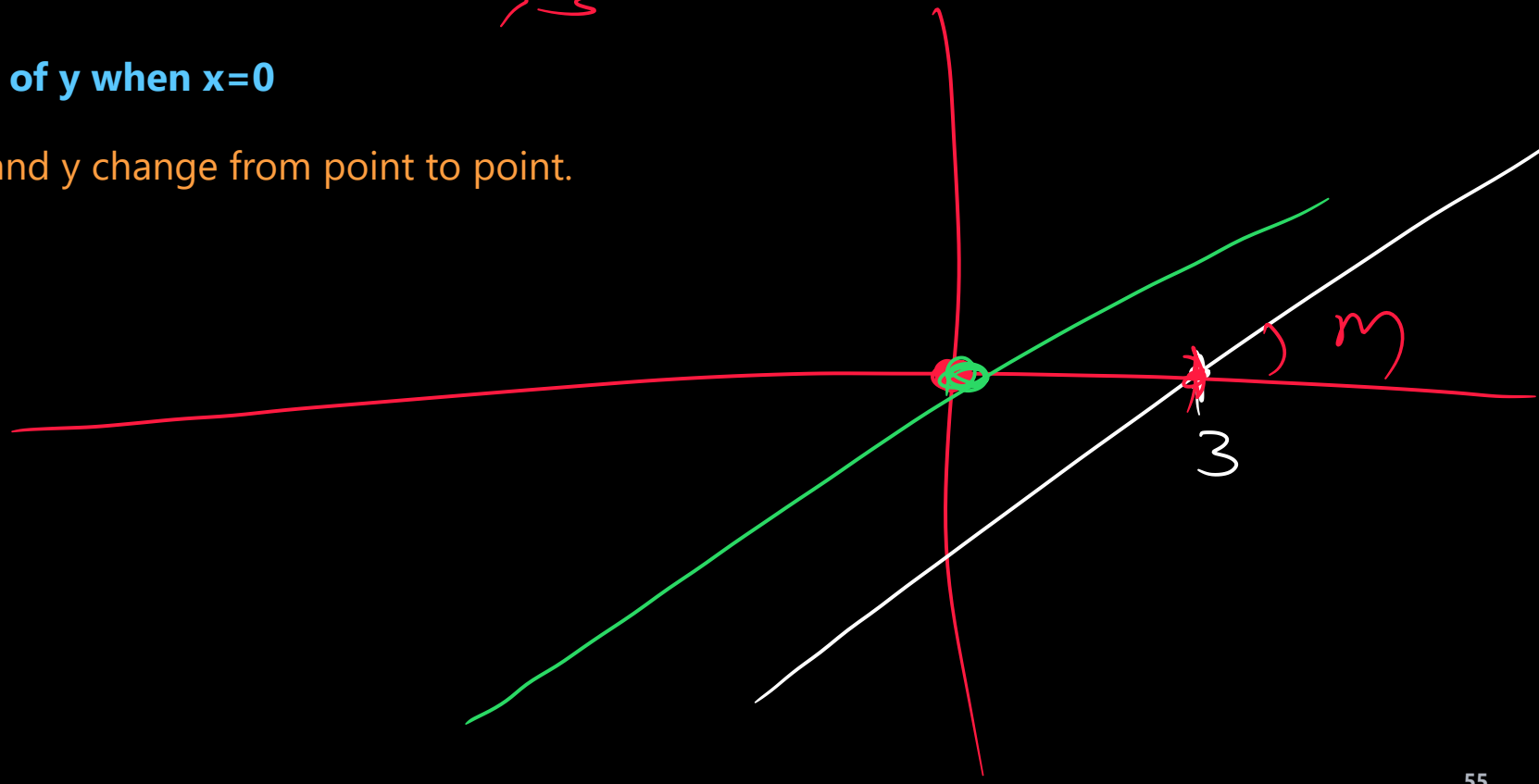
$$y = mx + c$$

m = slope **c = y-intercept = value of y when x=0**

m and c are fixed for one straight line. x and y change from point to point.

$y = mx + 3$

$y = mx + 0$



Read slope and y-intercept directly

First arrange the equation so y is alone. Then compare term by term.

y = (4/3)x + 4 → **m = 4/3, c = 4**

y = −√3 x − 4 → **m = −√3, c = −4**

The constant c is where the line cuts the y-axis, not the x-axis.

m = 4/3

c = 4

c = 4 and θ = 53°

A line cuts the y-axis at 4 and makes 53° with +x-axis. Write its equation and find y at x=3.

$$y = \frac{4}{3}x + 4$$

y at $x = 3$

$$y = \frac{4}{3} \times 3 + 4$$

$$= 8$$

Simple solution

Step 1: m = tan53° = 4/3 c = 4

Step 2: y = (4/3)x + 4

Step 3: at x=3: y = (4/3)(3) + 4 = 8

Equation: y=(4/3)x+4 y(3)=8

c = −2 and θ = 30°

A line cuts y-axis at −2 and makes 30°. Write its equation. Find y at x=2√3.

Eqn of Line

& y at $x = 2\sqrt{3}$

$$y = \frac{1}{\sqrt{3}}x - 2$$

$$y = \frac{1}{\sqrt{3}} \times 2\sqrt{3} - 2 = 0$$

Simple solution

Step 1: $m = \tan 30° = 1/\sqrt{3}$ $\quad c = -2$

Step 2: $y = x/\sqrt{3} - 2$

Step 3: $y(2\sqrt{3}) = 2\sqrt{3}/\sqrt{3} - 2 = 0$

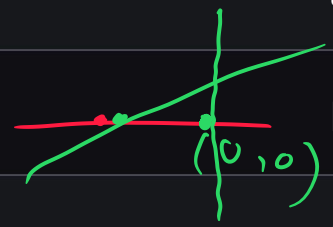
$y = x/\sqrt{3} - 2$ $\quad y(2\sqrt{3}) = 0$

Rough work / extra work

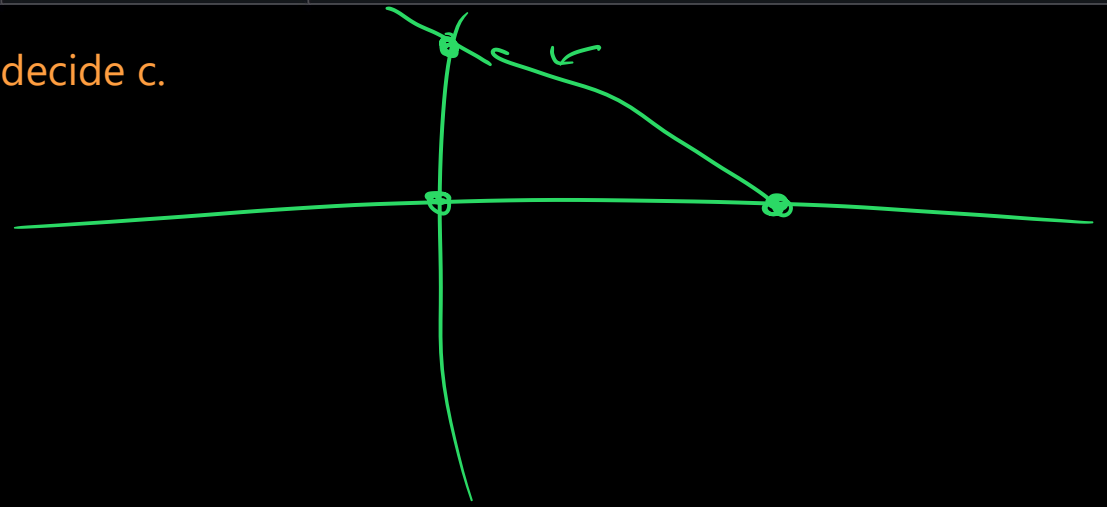
Signs of m and c — four quick pictures

line behaviour	m	y-axis crossing	c
rises, crosses above origin	+	above	+
falls, crosses above origin	–	above	+
rises, crosses below origin	+	below	–
falls, crosses below origin	–	below	–





Do not use where the line cuts the x-axis to decide c.



$c = 5$, $\theta = 150°$, $x = -4$

A line has y-intercept 5 and angle 150°. Write its equation and find y at $x = -4$.

$$y = \frac{-1}{\sqrt{3}}x - 4 + 5$$

Simple solution

Step 1: $m = \tan 150° = -1/\sqrt{3}$ $\quad c = 5$

Step 2: $y = -x/\sqrt{3} + 5$

Step 3: $y(-4) = 4/\sqrt{3} + 5$

$y = -x/\sqrt{3} + 5$ $\quad y(-4) = 5 + 4/\sqrt{3}$

$c = -4$ and $\theta = 120°$

A line cuts the y-axis at -4 and makes angle $120°$. Write its equation.

Simple solution

Step 1: $m = \tan 120° = -\sqrt{3}$ $\quad c = -4$

Step 2: $y = mx + c$

$$y = -\sqrt{3}\,x - 4$$

Rough work / extra work

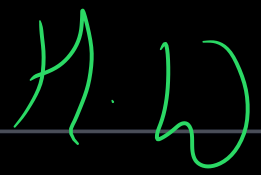
Horizontal and vertical line equations

horizontal through (0,4): m=0, c=4 → y=4

vertical through (3,0): x=3

A vertical line cannot be written as y=mx+c because its slope is not defined.

x-cut = −3, y-cut = 4



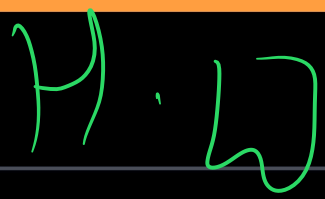
A line crosses the x-axis at −3 and y-axis at 4. Write its equation.

Simple solution

Step 1: $c = 4$ $\quad$ $m = \text{rise/run} = 4/3$

$$y = (4/3)x + 4$$

x-cut = 4, y-cut = 3



A line crosses the x-axis at 4 and y-axis at 3. Write its equation.

Simple solution

Step 1: $c = 3$ $m = (0-3)/(4-0) = -3/4$

$y = -(3/4)x + 3$

Rough work / extra work

Draw two straight-line graphs

P. 20

Draw $y = \sqrt{3}x + 4$ and $y = x$. First identify m and c for each line.

First ① m & c

Then, with ① & c

draw

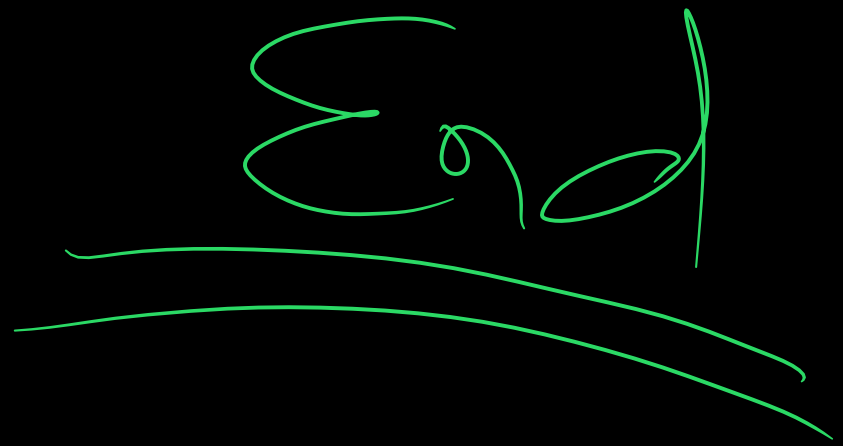
Draw two more straight-line graphs

Ex. 2

Draw $y = -x$ and $y = (4/3)x - 5$. First identify m and c.

Find intercepts before drawing

Draw $y=2x-1$ and $y=-2x+4$.



Coordinates, distance and slope

$$\text{distance}_2D = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2]}$$

$$\text{distance}_3D = \sqrt{[(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2]}$$

$$m = \tan\theta = \Delta y/\Delta x$$

rising → m positive falling → m negative horizontal → 0 vertical → undefined

Straight-line lock-lines

$$y = mx + c$$

m = slope **c = y-intercept**

x-intercept: put y=0 **y-intercept: put x=0**

parallel lines have equal slope

Slope is tanθ, not θ. A vertical line is x=constant.